

Su-un Lee

PhD Candidate at The University of Chicago

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Education

Ph.D. in Quantum Science and Engineering

Sep 2022 – Present

University of Chicago, Chicago, IL

Advisor: [Prof. Liang Jiang](#)

Anticipated graduation: 06/27

B.S. in Physics & B.S. in Mathematical Sciences

Mar 2016 – Aug 2022*

Seoul National University, Seoul, South Korea

Summa Cum Laude

* Military service [Mar. 2019 – Oct. 2020].

Work Experience

Returning Intern: Research Scientist (Scheduled)

Jun 2026 – Sep 2026

IBM Thomas J. Watson Research Center, Yorktown Heights, NY

Mentor: [Kunal Sharma](#).

Quantum Computing Theory Research Scientist Intern

Jun 2025 – Sep 2025

IBM Thomas J. Watson Research Center, Yorktown Heights, NY

Mentors: [Kunal Sharma](#), [Abhinav Deshpande](#), and [Kevin Smith](#).

Featured in [Engineering the summer: Advancing quantum computing at IBM](#)

Research Highlights

Opportunities and Limitations of Near-term Quantum Computers

- *Classical (in)tractability of quantum advantage experiments*: Identified the limitations of classical methods for several quantum advantage experiments, including advanced Pauli propagation and tensor network simulation.
- *Simulating dissipation with noisy ancilla*: Proposed a method to realize dissipative quantum dynamics with noisy ancilla qubits.

Information Processing and Complexity in Random Quantum Circuits

- *Classical simulation of random circuits*: Developed classical simulation methods for random circuits with generic noise, or noiseless shallow-depth random circuits.
- *Information processing in noisy random circuits*: Discovered a universal information spreading behavior that is intrinsic to noisy random circuits.

Benchmarking Near-term and Early Fault-tolerant Quantum Devices

- *Benchmarking logical quantum information processing*: Developed benchmarking methods for logical states and logical Pauli channels.
- *Benchmarking dynamical quantum circuits*: Developed methods for benchmarking quantum circuits with mid-circuit measurements and reset channels.

Undergraduate Research Program, Seoul National University

2020 – 2022

- Quantum information scrambling in continuous-variable systems: Advised by [Prof. Hyun-seok Jeong](#) and [Prof. Hyukjoon Kwon](#).
- Deep learning for dark-matter-only cosmological evolution: Advised by [Prof. Ji-hoon Kim](#) and [Prof. Myungjoo Kang](#).

Technical Skills & Experience

Classical Simulation of Quantum Systems / Quantum Circuits

- Developed a pipeline for Projected Entangled Pair States (PEPS) and Belief-Propagation (BP) simulations (being actively used in multiple projects in IBM Quantum).
- Implemented MPS/MPO simulation from scratch using only NumPy functions ([GitHub repository](#)).
- Large-scale stabilizer simulation for pure and mixed states.

Machine Learning and Deep Learning

- Construct and train deep neural networks with PyTorch and TensorFlow, accelerated with Multi-GPU programming.
- Developed automatic hyperparameter optimization routines with Optuna.

High-Performance Computing (HPC)

- Cloud computing and multi-GPU programming supported by NVIDIA Academic Grant Program.
- Midway, Research Computing Center, University of Chicago
- [Nuiron](#) & [Neuron](#), Korea Institute of Science and Technology Information

Programming

- Programming languages: Python, Julia, etc.
- Version control: Git and GitHub

Awards & Honors

- **Kwanjeong Educational Foundation Scholarship**

2022 – 2027

- **Eminence Scholarship**, Seoul National University

2016, 2018, 2019

Publications

In reverse chronological order. See also [Google Scholar](#)

- [1] William Kirby, *et al.*, Observation of Improved Accuracy over Classical Sparse Ground-State Solvers using a Quantum Computer. (2026), arXiv:2603.03496
- [2] **SL***, Ming Yuan*, Senrui Chen, Kento Tsubouchi, and Liang Jiang, Efficient benchmarking of logical magic states *Phys. Rev. Lett.* **136**, 050602 (2026).
- [3] Han Zheng, Chia-Tung Chu, Senrui Chen, Argyris Giannisis Manes, **SL**, Sisi Zhou, and Liang Jiang, Efficient learning of logical noise from syndrome data. (2026), arXiv:2601.22286 [quant-ph].
- [4] **SL**, Soumik Ghosh, Changhun Oh, Kyungjoo Noh, Bill Fefferman, and Liang Jiang, Classical simulation of noisy random circuits from exponential decay of correlation. (2025), arXiv:2510.06328 [quant-ph]. (Submitted to *Phys. Rev. Lett.*).
- [5] Yifan F. Zhang*, **SL***, Liang Jiang, and Sarang Gopalakrishnan, Classically sampling noisy quantum circuits in quasi-polynomial time under approximate Markovianity (2025), arXiv:2510.06324 [quant-ph].
- [6] **SL**, Changhun Oh, Yat Wong, Senrui Chen, and Liang Jiang, Universal spreading of conditional mutual information in noisy random circuits. *Phys. Rev. Lett.* **133**, 200402 (2024).

* Equal contribution

Presentations

Invited Talks

- Google Quantum AI Tensor Network Seminar, Google Quantum AI, CA Jan, 2026
- IBM Quantum Computing PIC Seminar Series, IBM T. J. Watson Research Center, Yorktown Heights, NY Oct, 2025
- IBM Quantum Computing PIC Seminar Series, IBM T. J. Watson Research Center, Yorktown Heights, NY Sep, 2025
- Quantum Information Science and Engineering Seminar, University of Chicago, Chicago, IL Dec, 2024

Contributed Talks

- Global Physics Summit 2026, Denver, CO March, 2026
- Korea Quantum Information Science Conference (K-QIS), Seoul, South Korea Nov, 2025
- Global Physics Summit 2025, Anaheim, CA March, 2025
- APS March Meeting 2024, Minneapolis, MN March, 2024

Academic Service

Program Committee

- *2nd Workshop on HPC/AI Integration with Quantum Computing/Networking (HAIQ 2026)*

Journal Referee services

- *Physical Review A* (5 times)
- *PRX Quantum* (2 times)
- *Physical Review Applied* (2 times)
- *Quantum* (1 time)

Conference Reviewer

- *Theory of Quantum Computation, Communication and Cryptography (TQC 2026)*
- *29th Annual Conference on Quantum Information Processing (QIP 2026)*
- *25th Asian Quantum Information Science Conference (AQIS 2025)*
- *Theory of Quantum Computation, Communication and Cryptography (TQC 2025)*
- *28th Annual Conference on Quantum Information Processing (QIP 2025)*

Teaching Experience

University of Chicago

- Guest Lecturer – Advanced Quantum Information and Computation Winter 2025
- Teaching Assistant – Advanced Quantum Information and Computation Winter 2025
- Teaching Assistant – Intermediate Quantum Engineering Autumn 2023

Seoul National University

- Teaching Assistant – Foundation of Physics 1 Spring 2022
- Teaching Assistant – Foundation of Physics 1 Spring 2021
- Teaching Assistant – Basic Calculus 1 Spring 2018

References

Liang Jiang

Professor of Molecular Engineering
University of Chicago
Email: liang.jiang@uchicago.edu

Bill Fefferman

Associate Professor of Computer Science
University of Chicago
Email: wjf@uchicago.edu

Kunal Sharma

Senior Research Scientist / Team Manager
IBM T. J. Watson Research Center
Email: kunals@ibm.com